Lecture notes not to substitute the book !!!

The rock containing a higher density of pre-existing microcracks will generally have a lower peak strength, assuming the cracks have comparable sizes, orientations and frictional properties.

The explanation involves three mechanisms.

1. Microcracks act as stress concentrators

Under compression, favorably oriented pre-existing cracks can undergo frictional sliding, producing tensile stress concentrations at their tips.

These concentrations can initiate wing cracks when the mode-I stress intensity factor reaches the fracture toughness:

$$K_I \geq K_{Ic}$$

For a comparable geometry, larger flaws produce greater stress intensity under the same driving stress. A greater number of cracks also means more potential sites for damage initiation, although crack density alone does not determine the stress intensity at an individual tip.

2. Cracks interact and promote further propagation

As wing cracks grow, their local stress fields begin to interact.

In a rock containing many closely spaced cracks, growing fractures are more likely to encounter neighboring defects. These interactions can promote further propagation and crack coalescence.

However, interactions do not always enhance crack growth: depending on crack geometry and orientation, neighboring fractures may also shield one another.

3. A connected fracture network develops more easily

When growing cracks connect, deformation begins to concentrate into a smaller region.

The formation of a connected damage zone can progressively reduce the rock's load-bearing capacity, eventually leading to macroscopic failure.

A higher initial crack density can therefore allow the transition from distributed damage to a localized fracture network at a lower applied differential stress.

An important distinction:

The effect of crack density should not be confused with the effect of confining pressure.

- Confining pressure tends to suppress frictional sliding and tensile wing-crack opening, increasing the stress required for brittle failure.
- Initial crack density influences the number and spacing of possible fracture initiation sites and the likelihood of crack interaction and coalescence.

Both affect peak strength, but through different aspects of the micromechanical process.

There is also a qualification: **more pre-existing cracks do not necessarily imply lower strength.** Crack length, orientation, spatial distribution and interactions are important. Indeed, the micromechanical models discussed by Paterson and Wong show that fracture geometry and the evolution of damage must be considered together.

The central takeaway for Lecture 13:

A rock does not fail simply because individual microcracks begin to grow. Macroscopic failure develops when the evolving population of microcracks interacts, coalesces and progressively localizes into a connected fracture zone.

This is why two rocks with identical mineralogy and fracture toughness can exhibit different peak strengths: their initial microstructures determine how damage develops under loading.

Why confinement raises peak strength

First: What prevents the parent flaw from sliding?

Increasing \(\sigma_3\) raises the normal compression on the inclined flaw, increasing frictional resistance. At a fixed differential stress, the available shear traction remains unchanged, so sliding becomes less favorable.

Second: What prevents the wing crack from growing?

The same confinement also opposes tensile opening, reducing the effective mode-I driving force at the wing-crack tip. Additional axial loading is therefore required to initiate or extend the crack.

This is the bridge between frictional activation, fracture toughness and pressure-dependent rock strength.

4. The physical connection: pressure sensitivity, flaw orientation and stress intensity factor

A. Remote stress determines traction on an inclined flaw

Consider a closed microcrack inclined by an angle(β) to the maximum compressive principal stress (σ_1), under axisymmetric compression with ($\sigma_1 \geq \sigma_3$).

Taking compression as positive, the normal and shear tractions are:

$$\sigma_n = \sigma_1 \sin^2 \beta + \sigma_3 \cos^2 \beta$$

$$|\tau| = (\sigma_1 - \sigma_3)\sin \beta \cos \beta$$

The orientation of the flaw determines how the same remotely applied stresses are resolved into forces tending to slide it and forces clamping it closed.

This immediately explains why pre-existing flaws do not all activate simultaneously.

B. Sliding depends on shear traction minus frictional resistance

For a cohesionless, closed flaw, the condition for frictional sliding is:

$$|\tau| \geq \mu \sigma_n$$

A useful measure of the tendency to slide is therefore

$$T_{\text{drive}} = |\tau| - \mu \sigma_n$$

The orientation maximizing this quantity satisfies:

$$\beta_{\text{opt}} = \frac{1}{2} \tan^{-1}\left(\frac{1}{\mu}\right)$$

For (μ=0.6), this gives approximately (29.5°) to (σ_1).

There is an important nuance: this optimum describes frictional activation in the idealized sliding-crack model, not necessarily the orientation of the macroscopic fault eventually formed.

C. Sliding generates local tensile stresses and wing cracks

What does the stress intensity factor physically mean?

1. Why is stress amplified near a crack tip?

Imagine a rock containing a pre-existing crack subjected to an external stress.

If the material were homogeneous and free of cracks, the stress distribution would be relatively uniform. However, a crack interrupts the transmission of forces through the material. The stresses must redistribute around its tips.

As a result, the local stress near a crack tip can be much greater than the externally applied stress.

In ideal linear elastic fracture mechanics, the near-tip stress field has the form:

$$\sigma_{ij}(r,\theta) \approx \frac{K_I}{\sqrt{2\pi r}} f_{ij}(\theta)$$

where:

- (r) is the distance from the crack tip.
- (θ) is the angular position around the tip.
- ($f_{ij}(\theta)$) describes the angular distribution of stress.

- (K_I) is the mode-I stress intensity factor.

Physical interpretation: (K_I) describes the intensity of the opening-mode stress field concentrated near the crack tip.

The larger (K_I), the stronger the local opening stresses at a given distance from the tip.

The mathematical expression predicts that stress increases as we approach the crack tip. The apparent singularity at (r=0) is an idealization: real rocks develop nonlinear deformation, microcracking and other processes that regularize the stresses at very small scales.

2. Why does crack length matter?

For a simple tensile crack of half-length (a) in an elastic material:

$$K_I = Y\sigma\sqrt{\pi a}$$

Here,(σ) represents the remote tensile stress opening the crack, and (Y) is a dimensionless factor accounting for geometry.

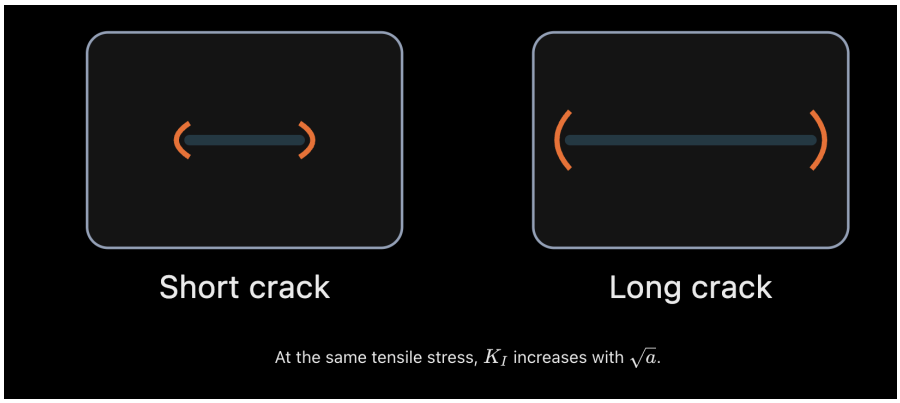
This equation illustrates two important concepts.

First, increasing the applied stress increases the intensity of the stress field near the crack tip.

Second, even at identical applied stress, a longer crack produces a greater stress intensity factor.

That is why the size of pre-existing flaws influences fracture initiation and rock strength.

Illustrative example: same tensile stress, different crack lengths



3. What is the difference between (K_I) and (K_{Ic})?

This is the distinction I would emphasize most strongly.

Stress intensity factor (K_I)	**Fracture toughness (K_{Ic})**
Describes the crack-tip loading	Describes the material's resistance to crack extension

Depends on applied stress, crack geometry and crack length	Depends on material properties and environmental conditions
Changes as the sample is loaded and the crack develops	Is often treated as a material parameter under specified conditions

Both have units of ($\mathrm{MPa}\sqrt{\mathrm{m}}$).

A useful analogy is:

- (K_I): How strongly are we driving the crack to open and propagate?
- (K_{Ic}): How much crack-tip loading can the material resist before the crack extends?

When the driving force reaches the resistance:

$K_I \geq K_{Ic}$
mode-I crack extension becomes possible.

Reaching this criterion does not necessarily imply catastrophic failure of the entire rock. A crack may initiate and subsequently propagate stably or arrest as the stress field and geometry evolve.

4. But our rock is under compression. Where does mode-I loading come from?

This is where we connect fracture mechanics to the sliding wing-crack model.

The externally applied principal stresses, (σ_1) and (σ_3), are compressive. However, the rock contains inclined pre-existing flaws.

When the resolved shear traction becomes large enough to overcome frictional resistance, the faces of a favorably oriented flaw can slide relative to one another.

That sliding redistributes stresses around the flaw tips and can generate local tensile stresses, even though the remote stress field is compressive.

These local tensile stresses drive the opening of wing cracks that tend to curve toward the maximum compressive stress direction.

The mode-I stress intensity factor at a wing-crack tip is therefore governed by the combined effects of sliding on the parent flaw, the remote stresses, and the evolving wing-crack geometry.

Increasing confining pressure (σ_3) generally suppresses this process by increasing frictional clamping of the parent flaw and opposing wing-crack opening.

The important insight is that (K_I) is not simply equal to the remote compressive stress multiplied by a crack-size factor. The simple equation in Section 2 describes a tensile crack; the full sliding wing-crack problem has a more complex stress-intensity solution.

5. How does this connect to macroscopic failure?

A single wing crack reaching (K_{Ic}) does not mean the rock has failed.

As loading continues, multiple cracks can grow, interact and coalesce. Their interactions modify local stress intensity factors, sometimes amplifying crack growth and sometimes shielding neighboring cracks.

Eventually, the development of a sufficiently connected and localized fracture network may lead to macroscopic instability.

This is the essential transition developed in Paterson and Wong's discussion of the micromechanics of brittle fracture.

Once the parent flaw slides, its tips experience strong, heterogeneous stress concentrations. These can generate a local opening-mode stress intensity factor (K_I), even though the remote principal stresses are compressive.

A tensile wing crack can propagate when

$$K_I \geq K_{Ic}$$

where (K_{Ic}) is the mode-I fracture toughness.

In a simplified dimensional description:

$$K_I \sim T_{\text{drive}} \sqrt{\pi c}\, f(\text{geometry})$$

where (c) characterizes the parent-flaw half-length. This is a scaling illustration, not the complete wing-crack solution: the actual (K_I) also depends on the wing geometry, its growth, crack interactions and the confining stress.

This explains why larger flaws can generate higher stress intensities under comparable loading, and why the geometry and orientation of pre-existing defects matter.

D. Why does peak strength increase with confinement?

Increasing (σ_3) affects the system in two related ways.

First, increasing normal compression makes frictional sliding more difficult. Second, confinement suppresses the opening and extension of tensile wing cracks.

Consequently, the applied axial stress required to initiate and propagate a population of microcracks tends to increase.

However, peak strength is not simply the stress at which the first wing crack appears.

As described by Paterson and Wong in Chapter 6, the later evolution of damage involves growth, interaction and coalescence of many cracks. In the Ashby–Sammis approach, macroscopic instability emerges from this collective damage evolution.

The physical progression is therefore:

Remote principal stresses (σ_1, σ_3)

Orientation-dependent traction (τ, σ_n)

Sliding and local stress concentration

Wing-crack nucleation and growth ($K_I \geq K_{Ic}$)

Crack interaction, coalescence and localization

Macroscopic peak strength and instability

This is also how I would interpret Figure 50 of Paterson and Wong: it links the experimentally observed failure envelope to micromechanical properties such as friction coefficient, initial damage and fracture toughness.

The model is especially useful because it explains why initial crack density affects the strength of a rock even when its mineral composition is unchanged.